A10.2 Major interface categories

The following sections describe the major interface categories.

A10.2.1 Read/write interface

A read write interface includes the following AXI channels:

AR Read address channel.

R Read data channel.

AW Write address channel.

W Write data channel.

B Write response channel.

A10.2.2 Read-only interface

A read-only interface supports only read transactions and includes the following AXI channels:

AR Read address channel.

R Read data channel.

Note

A read only interface does not support exclusive accesses.

A10.2.3 Write-only interface

A write-only interface supports only write transactions and includes the following AXI channels:

AW Write address channel.

W Write data channel.

B Write response channel.

Note

A write only interface does not support exclusive accesses.

A10.2.4 Memory slaves and peripheral slaves

AXI slaves are classified as Memory slaves or Peripheral slaves.

A memory slave must handle all transaction types correctly.

Peripheral Slaves are expected to have a defined method of access that establishes the types of transaction that can be used to access a device, and if there are any restrictions on how the device is accessed. Typically, the defined method of access is described in the data sheet for the component. Any access that is not a defined method of access might cause the peripheral slave to fail but is expected to complete in a protocol-compliant fail-safe manner, to prevent system deadlock. Continued correct operation of the peripheral slave is not required.

Because a peripheral slave is required to work correctly only for its defined method of access, a peripheral slave can have a significantly reduced set of interface signals.

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